

ActInf Textbook Group ~ Cohort 5 ~ Session 19 (Chapter 8, Part 2) ~ 3/11/24

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 19 (Chapter 8, Part 2) ~ 3/11/2024 | 58:50

all right welcome cohort 5 it's March 11th and Andrew thanks for taking us on to chapter eight part two so go for it all right okay it's our second week on chapter eight um and just a brief refresher that this chapter is in the second part of the text book and so we're now focusing more upon building our own models and getting a sense of how to draw relationships between phenomena of interest and how to actually map them to variables in active inference models chapter 7 even we saw um discrete State space models and so we were looking at pdps and now chapter 8 now transitions to continuous State space models which is sort of the the foundation uh the starting point for uh historically for developing active inference models a lot of the earlier work of uh Friston and others was on continuous models first um so here we're looking at continuous State space models which are well suited for modeling let's say physical fluctuations uh impinging on sensory receptors and for the continuous motion of the effectors or muscles we use to change the world around us is a quote from the chapter um here we have model examples for dynamical systems involved in motor control as well as the concept of generalized synchrony with the bird song example and finally constructing

Hybrid models uh near the end which would actually combine discrete and continuous variables um so it kind of almost add mixture of what we confronted in chapter 7 now with continuous variables being situated as well for example at a lower level of the model um since it's our second week uh from there I think we could just jump in does anyone have any thoughts uh on this chapter any questions anything they're they're still kind of uh crunching through uh in their mind about about how a lot of this

works

[Music]

all I'll at one point thank you Andrew for the quick opening I think it's an interesting point that like you said it was

initially uh with the earlier FR and Par models in continuous time and now the continuous time is being Revisited in the basic mechanic setting so it's kind of like a purely dynamical systems

justification which is very similar to the SPM textbook so kind of classical equations of motion which kind of turns out to be like the um classical Newtonian basic

mechanic so the kind of pre uh continuous time which is where most of the this chapter's

Focus um like including just the differential

equations including the

chaotic deterministic systems that's

just the kind of pure dynamical
systems approach again very similar to
SPM and then this textbook doesn't
really go into basing mechanics because
this textbook was published March 2022
so probably was written um
between First's

2019 free energy principle for a
particular physics and then obviously
the book being published in
so it's

mentioned

but many of the works that that
made it clearer are only after the book
just to kind of give a little more
context on the continuous
time

yeah so we start with continuous we move
to discrete and now we're back to
continuous but it's kind of using what's
been developed since then and also being
able to yeah combine the two um as
well

yeah in the the nested hybrid model at
the end of chapter 8 is basically like
picked up on by this
paper which we discussed in live stream
46 and they basically pick up on that
exact hybrid model I don't know if they
use a

figure let's see the hybrid

86 on page 166 oh sorry sorry yeah yeah
you're looking at the paper yeah just
that but I don't know they didn't
include it there but
uh

here yeah figure 8.6 so here continuous
time on the bottom

because of the D
derivatives and then discrete time on
the top because of the transition Matrix
and the explicit
time and then in this paper they say
okay with a belief desire intention like
BDI model including discussion of like
whether that is f psychology etc etc but
within that commonly used like can you
talk about beliefs desires and
intentions and all these
things they it it's not that it does
better or any different even per se it
just it's non-contradictory
so that's kind of an interesting first
way to put
it hey
oi is that particular paper does it
discuss much
like well I'm not sure how deeply you
want to go into that or maybe another
example but it it seems like a maybe a
nice way of like finding um for week two
here like um another application of what
we find in chapter 8
like what's what's going on in that
paper as far as like
distinguishing discrete from continuous
variables to have a recollection of like
the model set up there
let's look at another time but I think
they all they mainly explored like
differences in the belief desire
intention map into the model and not
sure how much they focused on that
sensory motor part but yeah gotta
sure yeah BR uh yes uh sorry uh could
you please go to that uh sorry that

desire part which they're mentioning uh
in that paper the desire thing yeah can
you please spend some time on
this yeah I think in

the you noticed it in the um the figure
probably see yeah D AI they're using for
discreet and that um

M AI I think for motor

AI so here they're looking at the um C
preference

distribution which is a distribution
over

observables and then um contrasting the
observation of tasting ice cream and not
tasting ice

cream so these two um alternative ways
you could parameterize that like in a
program or an equation here's 7.3 like a
coin flip 7.3 70% of the time I expect
to be tasting ice cream here it's 99%
expecting to be tasting ice

cream so then that c is
stronger and so it entails a stronger
motivational Force Through pragmatic
value in

E so that's one way to kind of look at
it and then there's the other the way
that the textbook kind of talked about
it is with the c

scaling but so depending on the program
sometimes you're working in the scaled
c um or unscaled scaled meaning to the
statistical distribution or like here so
that here Ice Cream versus no is being
described summing to one like a coin
flip like a real

distribution so then you get

or that could be represented as like 1

million to

one so just depends on exact situation

whether you're working in the kind of

setting where you can learn by counting

through experience or do other

modifications and then when that's used

as a distribution for pragmatic value

it's transformed into something sus to

one could you please send a link to this

p paper uh in the chat

box yeah it'll just be on the chapter 8

page yeah I'll put there

too yeah so if there are no

um anyone please feel free to bring up

any other questions you might have along

the way or you know toggle the reaction

to raise your hand

um but so in in this chapter along the

way we're we're introduced to things

like

attractors um we're introduced as well

to the concept of generalized synchrony

comes up a bit later

um and that's a particular notion that

um you know as we know active inference

agents are attempting to model the

hidden States uh of the environment so

they're attempting to model their

environment what's going going on around

them generaliz synchrony is this notion

that

um that the agent is basically their

internal states are modeling or in

certain ways beginning to synchronize

with um the states around them we have

no proof of course like the agent itself

has no

pure objective proof that it's somehow

perfectly understanding the environment and all variables around it but nonetheless there's a kind of synchronizing going on and this can be applied to a kind of multi-agent setting and so here uh later on in the chapter we're given this like bird song example which um you know the textbook is published in 2022 but the bird song example goes back to um at least 2015 potentially sooner um but there are a couple 2015 papers by Friston and Parr where basically investigating using continuous variables um they simulate uh two song birds singing songs right and what they do is that um there's a we start to see this kind of emergent phenomena of course with with active inference models one of the big uh aims here of doing modeling is to like set up your variables set up your model run a simulation and then see what kind of emergent phenomena arise with the bird song example they first um set up two birds that have identical models so not too not too uh surprising to find okay they have identical models they have the two song birds like quote unquote within earshot like they can the two song birds have actions where they can sing and then they can take in sensory observations that being of continuous auditory signals and so it's like the birds are singing to each other and listening to each other the emergent phenomena

that begins to happen is we start to see them taking turns it's as if they're singing from the same hymn sheet or the same script um and the generalized synchrony interestingly even occurs whenever they set up the model um the the simulation a second time around where the birds actually have different starting models so the birds start in different spots as far as their like State space like coordinates go and over time the bird's models actually begin through perceptual learning uh to to mirror one another so even though we start off with two different agents they start to synchronize um now they're singing the same him sheet again even though they're two you know very different birds as far as starting points go um the way that's modeled is through using attractors and La system and so uh in this case there's something that's called like a a synchronization of chaos which is rather interesting because you know with a chaotic system if you have two chaotic systems presumably they're you know even if you put them at the same starting point um their trajectories will very quickly diverge um you know with INF infin infinitesimally small chance that they will just magically line up but whenever you have two agents who are able to act um and listen uh to one another and they start to do this thing where they're following

the same Hy sheet suddenly you have a kind of synchronization of these two chaotic systems and we see those kinds of um Dynamics Illustrated in the in the chapter is kind of orbiting in figures 8.4 and 8.5 and um so yeah kind of a a lining up between different agents I think it's interesting for anyone who's curious about trying to set up systems or do multi-agent settings where they're trying to see you know how a joint action begins to play out between agents saying uh how agents can work together to um ultimately minimize their own distinct you know free energy um but by in order to do that they actually have to like yeah entrainment is a good word like one of them entrains the other um so that they can stay on the same page minimize prediction errors and it's fascinating because the there are all kinds of uh Rich behaviors and dynamics that that occur um like the the the songs that are being sung presumably they like vocal Glides and brief pauses and so it's not just you know super simple Behavior that's that's being synchronized here but nonetheless it occurs see the bird song also I I like how this chapter brings up oh uh Peres is that your hand up again uh yeah sorry I have a I may be wrong actually but just correct me um so for synchrony there

needs to be some kind of a coupling uh
I'm just wondering is it anywh Ed with
the generative process and the
generative models like the act like we
have the like the Gen process is in a
way uh complementary to the generative
model is it possible that that plays an
impact sure so so in this
example there's not too much going on in
terms of U so just as a starting point
so with modeling you're going to be
defining a generative model and you're
going to be defining a generative
process in this example there's not too
much distinct going on whenever a
generative process is defined in fact
here it's like we have two generative
models so two
agents and they're coming into contact
with each other so it's like for bird
number one
its model is itself um its process that
it's coming into contact with is just
the other bird so you can think about
the the observations that bird number
one is taking in they're all being
generated by by bird number two um as
part of the the environment the process
that's around it um does that make
sense I think you you're right I think
instead of thinking terms of a
generative process Alto together like
two different agents and they have like
uh uh similar kind of internal model and
uh and that is kind of causing kind of
some kind of a coupling mechanism and so
over a long period of time uh they may
start to have behaviors which are

similar to each other no like you mentioned that him sheet right so one person may say the starting lines and then then after sometime the second person will say the second thing then the third will say first will say the third and like that observations might happen yeah it's like the pendulum example which I think uh was mentioned last week like we have these two pendulums which are swinging and over a very very long period of time they'll they'll kind of um oscillate together no yeah yeah know that's exactly right and uh that's that's also referenced in the bird song paper so yeah just basically being able to start two different in that yeah in that case your pendulums are kind of like your agents um and yeah via you know via physics via the the the general logic that we're looking at here of like attractors yeah they begin to kind of move towards attracting points but they're acting as each other's attracting points so to speak um such that yeah their their behaviors start to to synchronize and so with the the birds the bird song is an interesting example versus the pendulums because the pendulums they just like continue moving for the birds um we have this turn-taking Dynamic that begins to occur because the the birds are trying to in terms of you know act act of inference and the free energy principle they're actively Trying to minimize their prediction error and the way they do that here is that they want to hear a

song like they want to take
in uh sensory observations that show
that are realizing the preference of
hearing a a song and the two ways they
can do that that of course our
perception changing their mind their own
mind or uh action which is changing the
world and so this turn taking Dynamic
involves a combination of sensory
attention and sensory
attenuation uh is how it's phrased here
and these are also um interesting
mechanics that we find in in just about
any active inference model where action
is involved um you have to so the birds
take turns because like how do you keep
a song going well one uh you need to
hear that it's going on so they
attenuate or or kind of like pause um
their
um their incoming sensory
signals uh to uh commit actions that
realize hearing the song it's like oh I
don't hear a song I need to sing that's
that's realizing my preference um okay
I've sung um now I
pause I hear the other bird singing I
take that in as a sensory signal the
other bird uh begins to
sing maybe it sings notes that I expect
great maybe it sings notes that I don't
now we have prediction error starting to
to jump up yeah and so whenever that
happens of course the bird number one
has to minimize its prediction error and
so it starts to sing itself again and
because these two birds are basically
doing this back and forth there's a kind

of synchrony where they're because they're both trying to minimize the prediction error the way that they sing the actions that they take the specific ways that they sing um like they start to synchronize they start to to match notes start matching expectations and that's the kind of thing that we see playing out in the the graphs um and the figures in the in the and the textbook um especially figure 8.5 on the right the authors here you know frisen and everyone they're trying to show uh like a before learning and after learning for these two birds on the left- hand side it's like okay these are the these are the Dynamics playing out here are the here are the dynamical equations um and then on the right you can see the upper graph before learning these are kind of like the state spaces of each Bird's expectations they're they top- down predictions um they're a little bit more all over the place whenever they start they begin to synchronize so that you know whenever whenever bird want and recall these are continuous um State spaces which is why we are able to do this as coordinates rather than having to look at like discreet um you know Point masses or something um so where bird number one's uh or excuse me yeah the first Bird's expectations in that lower right graph whenever they're around 10 um you know whenever we whenever we plot them um the second Bird's expectations are also around a value of 10 whenever bird

number one's uh expectations or predictions are around 50 same case for the second bird so that's that's what they're trying to get at with these graphs is yeah they begin to the expectation even though they're dynamically shifting around we're never always at zero we could be at 50 this moment we could be at 5 this moment um the the important point is that the first Bird's predictions and the second Bird's predictions are um basically both lining up at just about any given moment one one interesting thing I hadn't seen or before was like here it's going between at least 60 but here it's only going to 40 yeah presumably I mean we have before learning and after learning so presumably they maybe just before learning haven't explored quite as much of the this state space yet right like we have this much more EXP exploration after learning you've run more you know trials or at least extended the the time Horizon and then here they are like starting to syn up singing new notes that they'd never sung before or however you'd like to phrase that and yeah the the scale starts expanding because they're they're trying new things not sure nice nice interpretation yeah it's a it's it's a thought in any case and I also noticed it looks on the after learning graph it

it looks slightly more chaotic or just
messy not quite as synced up once you
get past 50 on the after learning my
guess is that like oh this is a new the
new terrain that they're exploring here
um yeah these little micro little micro
densities
yeah
so presumably after more trials like
we'll see an even
finer kind of line or syncing up what
happens between
them we should find the at lab it might
it may even be an
SPM
example there's also a kind of mini
Standalone
SPM like while it is a mat lab package
there's a a kind of like
compiled Standalone
SPM oi or or others people like it it
it's worth a a try yeah and the bird
song I I I could check or we could
check together because it's kind of been
around long enough to probably be
included in
the Dem
demo yeah actually that's that's right
um I was just having a look at the the
paper um I think I dropped in the chat
there but yeah just just about any of
these uh friston co-authored papers
prior at least prior to maybe the past
couple years it's like going to a paper
crlf SPM see if they you know have a
script there um that they're you know
they'll always mention in the paper of
course there you go and um yeah it looks

like it is in there

um yeah for For Better or Worse of course matlab still requires um a license to use but but I myself have downloaded the the SPM Standalone before um you just you know unfortunately of course you cannot run M Matlab scripts if you don't have Matlab so spm's nice as far as like you get

this um this like graphical system all of these different options of demos you can click you can look at the graphical outputs and yeah it looks like this one um is included in there it's like that um bird song

umcore um oh no dmore demoore d.m so it can run M scripts in the background you just can't hop in and access them and modify them yourself uh to tweak the parameters or anything like that unfortunately but yeah it's nice it's there there like literally dozens of of preloaded scripts in there if you just download the SPM Standalone you just click one and uh you can you can watch what kind of graphical outputs it comes up

with yeah and we can also explore bringing it into RX and F and everything

yeah yeah

definitely

um and for anyone here who's not familiar um The Institute has an ongoing project Thursday mornings well sorry I'm in the US some ADV to that but um early on

Thursdays uh there's an RX infer project

for anyone who's interested in adapting the logic of active inference that we're reading about here to direct application in uh the Julia programming language and um right now we're going over different examples that are already provided by the developers byas Labs uh of this RX infer package um but um we'll be expanding from there uh different people you know are proposing their own Community projects uh you know diff different kinds of use cases are coming up with and that we could try uh emulating as a group uh and then another focus is that we would like to be able to reproduce some of these existing uh um projects or or topics for example here we could try and take the take the mat lab script for the bird song example and recreate it in Julia and uh finally just final note um is the um Julia programming language as well as the way that bias laabs is developing the RX infer package it's just it's nice that it's it's using this more um kind of sophisticated but also highly efficient method for the computation uh and message passing as we saw in the textbook back around chapter 5 um to to really just like being a ble to scale up these kinds of processes and and being able to just basically better use your computational resources save on some time um efficiency all that good stuff we in the programming World care about um yeah Francis you have your hand up uh yeah I just wanted if there's uh

two questions actually firstly uh if there's likely to be any support for a a pmdp group because I'm already having to Learn Python I I'm not I can't really commit to learning a second programming new programming language from scratch at the same time uh so I've been kind of slowly working through the um that nice pmdp tutorial um uh book Running it on the um is it collab uh and uh the other question is um is there anything to be leared or is it not a really helpful parallel uh between the idea of the bird song example as a kind of autocomplete and how uh does it does active inferences uh success here compare it uh in any way to um llms uh success at uh GPT for the world's smartest AO complete thanks just to quickly address it so right now like we do not have any active PDP specific group we could we could do a similar kind of thing right now the RX and fur is really it's it is kind of interesting it's it's um just from knowing PDP to medium degree it is different syntactically however like it is it's very different and it's a more General

than
pmdp in several ways not to say that
that that one can't have other you know
things the other doesn't
have but just the that um RX and
fur pulls way back like to the coin
flip which was really helpful and then
then builds up whereas PDP wraps it
focuses on just kind of getting you to
the discreet time active agent PDP as
fast as
possible because that's that's like not
every use case but that's
like many use cases and then it can kind
of be
modularized from there but RX and fur
is it's a it's a very promising
effort a lot broader so I I think it's a
good way to go I'm kind of excited to
learn it a lot of is very good at
it other than the second piece about the
the um llms well
one we we um we haven't had a a streamer
discussion on with these authors but
this is one
interesting paper
recently um and then yeah like it's
almost like a question of what
what is perceived as a um Computing
Benchmark especially if other
considerations like interpretability or
energy it's
like is it a challenge is it a Computing
advance to open a larger file with a
computer with more RAM it's like yes but
maybe it's not proportionally or
differently a kind of
benchmark so then what what are

benchmarks for active models and like or or even kind of in another um setting like with the morph stream with Sarah Hamburg different guests it's been interesting to like hear how they talk about like the different characteristics of the biometric Computing system because they are not it's not simply just like the um CPU Ram descriptions thanks so much that's it looks like a really um interesting uh paper yeah and only six pages oh no comments of six pages yes yeah it's a sure it's definitely a starter so I and I think that they and many other people are interested and and a big um um like of course the models are different if you look at the Transformer architecture um or um but but there's some interesting with the state space models like the Mamba there's there's a few people like Matt Perkowski who have um talked a little bit about the relationship between the state spaced models and active inference so that may be actually a closer relationship but um just in terms of how the Transformers are different than active inference models I mean in the textbook group chap these chapters are showing us what the ACM model is you can then just look at the Transformer architecture and just see how they are different

architecturally but then um in in this paper they kind of highlight this kind of uh ongoing there there's various aspects to highlight there's the epistemic value there's the um reinforcement and but there's also a lot of overlaps as these are just large statistical models yeah oh thanks that's great oh and this is another good one where um they talk about how you in active inference you can construct the causal graph to be like that there is an uncertainty parameter or like a temperature parameter on some other parameter or there's like introspective access that using the um classic Lars live stream 28 figures with the meta awareness so if you construct the model to be this way then this parameter actually has that role in the system whereas if you ask a prompt no matter what you ask or how you ask it um to a a Transformer architecture and what is it like to be you and all these kinds of questions that people are very interested in it's only ever giving that semantic space fuzzy statistic of Words Answer so it could say I'm very sure that it's this way or very sure it's that way or I'm not sure or I feel this way when it's that way but those are all just words generated from texts um statistics of texts so they will never approach an

actual

introspection so here they raised the

possibility of the actual

introspection that's very interesting a

totally different author list

uh is can we have that link on the um uh

encoder as

well thank you so

much this is all interesting with the

comparison to llms because I've been I'm

no expert on that topic but I've been

looking into it more myself and yeah

there is this kind of

um there's this kind of notion that okay

llms are great as far as um you know

kind of per performance goes with with

accuracy and um and all of the

computation there nonetheless yeah major

the kind of criticisms and the sort of

um debate or contrasting between llms

and actm is just more generally like oh

it takes a ton of computational

resources to train in llm and then once

you're kind of finished training it it's

like okay well there it is it's not a

it's not conceived as an agent who's um

who's able to engage in action uh right

so like try and it it's like to the idea

would be you'd have to have an llm that

had the capacity to actually go and

explore information itself and then

presumably it' have you know from an

active inference uh take it would have

to have its own preferences of how to

you know that it wants to realize so to

think of C Matrix in the or C vector and

the PDP scheme or however you would like

to think of it and um so it's it's the

it boils down to the idea that an llm is passive um it can't go explore it can't update information and new you know expand its uh State space dynamically or or do things that maybe u a more active uh agent could um it's kind of like okay well once you've trained it like that's what you've got um nonetheless I'm very curious about the paper that yeah share because that looks like it it's it's actually saying no llms are not entirely passive but let's talk about it yeah that that that's all true and also what's really interesting though is the um I mean what what we might call ecosystems of shared intelligence or just the multi-agent prompting they like when you on perplexity AI or on um any internet enabled like cloud provider um or even if even in the uh local retrieval augmented generation where it's like you have a library of documents and then the it pre um assembles into the context window what's relevant like that's an information forging so this is kind of like um is it active inference like low road like has it been constructed according to chapter six using like forked code from an act in Freo or is it just being described as in in terms of having particular uh States like figure uh uh 9.1 with the kind of rat in the Maze like making a cognitive model of another language model that that doesn't mean it it will

be useful to do

that but these models are prompting each other in ways that makes them have epistemic value like they do with web searches even if

one language model locally doesn't have certain properties so there's a lot of um

interactions which is kind of only to be expected and even these other these systems are basically multi-agent models like the large ones that are closed

source

but not in the kind of spatial Mobility

way properly just like in the combining

inputs gotcha I was actually personally

not familiar with the fact that like

these are technically multi-agent setups

um that's really interesting and also

makes an immense amount of sense actually

um yeah yeah and they're working off of

one another another I'm not trying to

over relate but just given this is the

the textbook group of course it's like

a you know we have the bird song example

where the birds are trying to help each

other um so to speak uh not we can't

that's an interpretation to say that

they're trying to help each other but

the point being that they're both trying

to minimize their prediction errors and

so they're playing off of one another

sharing information with one another or

in this case a song guess with LLMs in

a certain

sense of course them being constructed

very differently we're talking about

this right now but they're kind of
working off of one another to yeah yeah
I think another good uh point there in
the context of building the models is
like it's not necessarily about building
like one Megatron agent that has every
property it's about like what things are
there and
then starting to describe their
interfaces which are usually less
ambiguous and then describing the things
and and their interfaces and
interactions then maybe all you need is
oscillator versus like a lot of times
the kind of speculation about very very
Advanced kinds of models but as we're on
RX and
fur
like- which is awesome too the um the
the the philosophy far
outruns simple
models so depending on what we're really
trying to explore just you know
describing and then having and then
having the modularity of the model so we
could start with a
pendulum but then know that the
interface would be described like in a
stable way in RX and
fur so that would have made large
progress
towards different kinds of phenomena
without necessarily getting caught up in
trying to locate it too early in a given
architecture for a single
agent
can I do one more question does this
idea of

the uh
synchronization
um refer back to the phrase you used
earlier about was it community of
something a
shared something yeah yeah I think
here's here's a good citation okay good
this one's it was on archive for some
time then it was published here
here's the ecosystems of intelligence or
shared intelligence and this kind of
lays
out I guess other people can also give
their summaries I see it it doesn't make
any technical advances or descriptions
there's only one
figure um it's just laying out this idea
of the
interacting things through message
passing this is the
figure here's some manifold and then
belief
updating interesting reading
format and then um then they just talk
about how message passing amongst
diverse intelligences in ecosystems of
shared
intelligence that's kind of the frame
then they describe these um stages of
development
kind of an interesting read in terms
of their timelines the scriptures
Etc wow yeah thank you very
much so it looks like in that paper
they're
describing uh like certain aspects of
one
agent's own

model act as points of
reference
for other agents and vice versa so that
you end up with this kind of like
Ecology of belief
sharing a kind of collective
intelligent intelligence resting on
shared narratives and goals
um again not to to
overrate everything here but with the
bird song example I had actually like I
this past week took some time to like
actually read that paper itself the bird
song One not just its briefer
description uh in the textbook and
there's a lot of there's a lot of
interesting things actually going on in
that 2015 bird song paper I in some ways
this might be just designing ecosystems
of intelligence from first principles
more than likely has some relationship
there because the bird song example it's
not just of course it's not just about
trying to talk
about distinctly birds and and song
birds and like you know uh as as
interesting as that could be from a a
biological or Zoological or otherwise
standpoint it's really trying to get at
something like how do
agents share information with one
another and kind of in Trin one another
how do we have how and why do we see
things like convergence of
beliefs um as far as biological
organisms go we can see a
synchronization of things beyond the
words um we can see like whenever two

people are talking to each other their
the rate of their speech tends to begin
to converge over time or or exhibit you
know near identical characteristics
um the kind of vocabulary that people
use tends to kind of sync over time as
well um and then yeah there's so there's
this kind of shared narrative a kind of
um you know from a seal agent
perspective uh the single agent is
starting to model um the behavior of the
world around it the process as well as
its own um two agents might start to
model each other and in a sense uh there
uh the bird song paper claims it's not
just that once you get to this
multi-agent setting where where the
agents are entraining one another and
starting to show identical behaviors and
internal model mappings and U modeling
of the world of each other uh it's not
just that they're modeling the other
Bird's Behavior or their own behavior
they phrase it as they're beginning to
like our Behavior like there's a kind of
collective behavior on the on the
horizon here that they're starting to
model um yeah I I just think that's a
very very interesting and potentially
one of the core aspects of trying to to
get into this kind of collective
intelligence synchronization kind of
logic here yeah
yeah great point I mean when it's on a
onedimensional manifold like in the
extreme case or even just as a heris
like if 80% of the variability is
captured by one number like you're on a

a one-dimensional manifold on the
two-dimensional space that's the that's
what a manifold is and then yeah you're
totally right this is not really a uh
it's not an Ornithology paper in the
traditional sense because it is about
hermeneutics and it is with this other
joint paper that they
reference that's like a a twin paper
and so it it's it's totally interesting
that it has a lot to do like you said
with
narrative and with uh
interpretation here they
um they take like a dynamical systems in
approach to
interpretation later that is also like
um expanded and complemented by the
quantum reference frame Chris
Field's syntax and semantics
work but this is like a a perception
action
grounded dynamical
hermeneutics so not addressing all
semantic or Semitic questions but that
but like how dynamical
systems can have General synchrony which
doesn't mean doing the same thing it can
mean it can mean Mutual information
increases
through
differences like or in um in the Chris
Fields setting like the the principle of
unitarity and like that systems um come
into entanglement with each other and
then that was discussed like does that
mean that they go to the same States
like no not

necessarily and I say as a counselor who goes into conversations with people where they know their world but maybe they're stuck and I don't know their world but I have models of how to get unstuck

this

is uh

wow um very very very interesting thank you yeah no super super interesting I mean I think I I think there's enough to go off of here at least theoretically that potentially one of the key um aspects

of Counseling and as well as other things clinical therapy and so on is that uh just basically a so-called talking cure or some s some such thing is for someone who's having uh kind of repetitive uh you know behaviors or thoughts or you know whether you want to view them from a symptomatic perspective in the cases of like I don't know depressive rumination or a variety of other kind of things that come up like in one sense from an active inference perspective it might be that someone is having something

like difficulty with belief updating and that by talking like actually just talking with another human being um you might see the person who's who's you know the client you might start to see them like you know not not to be too theoretical or kind of philosophical here but there's almost a kind of potential s thinking to some degree of what we might call their internal States

or their behaviors or otherwise with
that of the counselor or the the
therapist you know um whole different
set of questions and and research to be
done or research that's already been
done just try and claim like that that
can have a a long-term impact of course
like counseling or therapy like there
are regular sessions so you're getting
like repeated instances of these these
temporary sinking EP uh sinking Up
episodes going on and then another
aspect of course is that the counselor
and or therapist they don't want to
actually themselves imitate um the
behaviors being seen by the client they
want to help not match um but yeah just
you know just trying to extrapolate a
little bit here yeah that's why we have
a a fairly precise definition of empathy
as if we were feeling the same as our
client as if we were in their world but
not actually being in their
world I have to leave thank you so much
all right sure next
time for chapter nine thank you Andrew
so see you all next time bye take care